

YUKAI WANG

PhD Candidate | Cho Chun Shik Graduate School of Mobility | KAIST

yukai@kaist.ac.kr | yukiiwong.github.io | GitHub: [yukiiwong](#)

RESEARCH PROFILE

My research develops a site-centric perspective on traffic world models, extending my interest in autonomous driving to the interactions and safety of an entire traffic scene. Using drone-recorded trajectories, I study multi-agent dynamics, observation dependence, closed-loop error propagation, and whether forecasts support defensible decisions and safety assessments. The longer-term goal is a transportation research tool for evaluating traffic scenes and digital twins, not only a model for controlling one vehicle.

EDUCATION

PhD Candidate | 2023 - present

Korea Advanced Institute of Science and Technology (KAIST)
Cho Chun Shik Graduate School of Mobility | Advisor: Prof. Tiantian Chen

Master's degree | 2020 - 2023

Southeast University
School of Transportation | Advisor: Prof. Zhiyuan Liu

Bachelor's degree | 2016 - 2020

Southwest Jiaotong University
Mao Yisheng Honors College

DOCTORAL RESEARCH | KAIST, 2023-PRESENT

World models from drone trajectories

Ongoing

Develop graph-conditioned recurrent state-space models with neural ODE motion decoding; evaluate multimodal prediction and observation-grounded versus self-conditioned rollout on drone trajectory data.

Understanding why closed-loop rollouts fail

Under review · AAAI 2027

Design paired action/graph source-switch diagnostics and temporal controls to investigate distribution shift and early error compounding in autoregressive multi-agent rollouts.

What does missing-history robustness measure?

Under review · AAAI 2027

Distinguish input-level missing-history tests from latent-pathway diagnostics; compare forecasters under matched information and intervention conditions.

ProSafeAV: world models meet tail-risk modeling

Under review · IEEE T-ITS · Earlier conference work: TRB 2025

Study world-model-based reinforcement learning with traffic-conflict indicators and extreme value theory; develop and evaluate the approach in CARLA-based simulation. Conference work listed at TRB 2025.

YUKAI WANG | PUBLICATIONS AND CURRENT SUBMISSIONS

JOURNAL PUBLICATION

Ziyuan Gu, **Yukai Wang**, Wei Ma, and Zhiyuan Liu (2024). A joint travel mode and departure time choice model in dynamic multimodal transportation networks based on deep reinforcement learning. *Multimodal Transportation*, 3(3), 100137. doi:10.1016/j.multra.2024.100137

CONFERENCE CONTRIBUTIONS

Yukai Wang, Tiantian Chen, and Sikai Chen. ProSafeAV: Enhancing Proactive Safety Performance of AVs using World Model-Based Reinforcement Learning and Extreme Value Theory. *TRB Annual Meeting 2025*. [Record]

Yukai Wang, Tiantian Chen, and Kitae Jang. A Synergistic Approach to Real-Time Crash Risk Estimation at Signalized Intersections: Integrating Learning-Based Anomaly Detection with Extreme Value Theory. *Conference in Emerging Technologies in Transportation Systems (TRC-30)*, September 2-4, 2024. [Record]

MANUSCRIPTS UNDER REVIEW

Status as of 22 September 2026.

Crash risk estimation using auxiliary bivariate anomaly scoring and non-stationary bivariate peak over threshold modeling

Under review | Analytic Methods in Accident Research (AMAR)

Yukai Wang, Tiantian Chen, Yasir Ali, and Kitae Jang

Traffic-conflict risk estimation with anomaly scoring and extreme value theory; revised manuscript.

Proactive Risk Modeling for Autonomous Driving with World Models and Extreme Value Theory

Under review | IEEE Transactions on Intelligent Transportation Systems (T-ITS)

Yukai Wang, Yuhao Luo, Hongliang Ding, Kitae Jang, Sikai Chen, and Tiantian Chen

ProSafeAV: linking imagined conflict severity to policy learning; revised manuscript.

Where Closed-Loop Traffic World Models Break and When Correction Pays

Under review | AAAI 2027

Paired diagnostics for the source and timing of closed-loop rollout errors.

Missing-History Robustness Is Not a Single Number

Under review | AAAI 2027

Separating input-level missing-history tests from latent-pathway diagnostics.

The AAAI entries concern two distinct studies. Additional ongoing projects are listed separately on the following page.

YUKAI WANG | ONGOING RESEARCH AND EXPERIENCE

ONGOING RESEARCH

From predictive validity to decision utility

Ongoing

Build recording-disjoint, future-free candidate-ranking audits with fixed candidate banks, analytic baselines, leakage checks, rank fidelity, and reference-relative regret.

Safety assessment from forecasted trajectories

Ongoing

Develop leakage-controlled pipelines from predicted trajectories to traffic-conflict indicators, event-level diagnostics, and tail-risk analysis.

Behavioral validity and cross-site generalization

Ongoing

Investigate layered behavioral validity in calibrated microsimulation and cross-site evaluation of graph-based and autoregressive traffic models.

INDUSTRY EXPERIENCE

Data Analyst Intern | Shokz | April - July 2023

North America Marketing Department

SELECTED AWARDS

2021 | **Top 20, AI Competition.** RLChina Reinforcement Learning Summer School.

2019 | **National First Prize.** 14th National Competition of Transport Science and Technology for Students.

2019 | **Honorable Mention.** Interdisciplinary Contest in Modeling.

2018 | **National First Prize.** China Undergraduate Mathematical Contest in Modeling.

2017 | **National First Prize.** CSEE Cup National University Students Electrical Mathematical Modeling Competition.

METHODS AND TOOLS

Graph neural networks; recurrent state-space models; neural ODEs; autoregressive models; reinforcement learning; surrogate safety measures; extreme value theory; recording-disjoint evaluation and bootstrap diagnostics. Python, PyTorch / PyTorch Lightning, CARLA-based simulation, Git, and LaTeX.

My research agenda extends world models from ego-vehicle control toward site-level traffic understanding, safety screening, and digital-twin evaluation. These are research goals, not claims of validated real-world intervention effects. Research story and figures: yukiwong.github.io/research/.